

Air University



Network Security (Lab)

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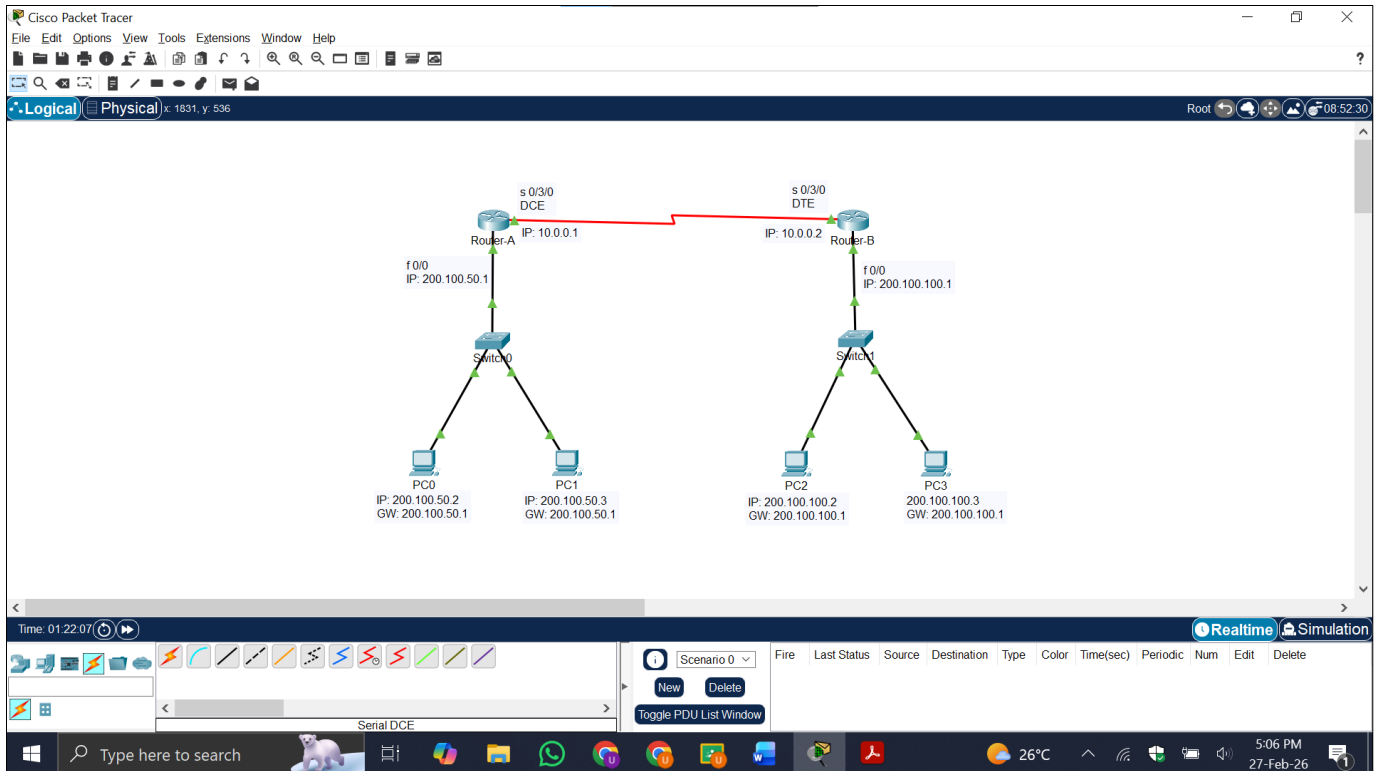
BS – Cyber Security – Fall – 24 (A)

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Lab Task # 03

“Static Routing and Telnet”

Topology



Static Routing

Configurations

Router – A

I completed the basic configuration of Router A for its serial and Ethernet interfaces.

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router-A
Router-A(config)#int s 0/3/0
Router-A(config-if)#ip address 10.0.0.1 255.0.0.0
Router-A(config-if)#encapsulation ppp
Router-A(config-if)#bandwidth 1000
Router-A(config-if)#clockrate 64000
Router-A(config-if)#^
% Invalid input detected at '^' marker.
Router-A(config-if)#clock rate 64000
Router-A(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router-A(config-if)#exit
Router-A(config)#int f 0/0
Router-A(config-if)#ip address 200.100.50.1 255.255.255.0
Router-A(config-if)#no shutdown
Router-A(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router-A(config-if)#exit
Router-A(config)#
```

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Router – B

I completed the basic configuration of Router B for its serial and Ethernet interfaces.

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int s 0/3/0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#encapsulation ppp
Router(config-if)#bandwidth 1000
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config-if)#exit
Router(config)#int f 0/0
Router(config-if)#ip address 200.100.100.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#
```

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Verification

I verified the configuration of both routers by using the command `show ip interface brief`.

Router – A

```
Router-A#show ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          200.100.50.1    YES manual up          up
FastEthernet0/1          unassigned      YES unset  administratively down down
Serial0/3/0              10.0.0.1        YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
Router-A#
```

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Router – B

```
Router(config)#hostname Router-B
Router-B(config)#exit
Router-B#
%SYS-5-CONFIG_I: Configured from console by console

Router-B#show ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          200.100.100.1   YES manual up          up
FastEthernet0/1          unassigned      YES unset  administratively down down
Serial0/3/0              10.0.0.2        YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
Router-B#
```

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Defining Routes

Configured static routes for both routers using the command '**ip route**', and confirmed the route's existence by using the '**show ip route**' command, where the letter S indicates that the static route has been successfully set.

Router – A

```
Router-A(config)#ip route 200.100.100.0 255.255.255.0 10.0.0.2
Router-A(config)#exit
Router-A#
%SYS-5-CONFIG_I: Configured from console by console

Router-A#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
    C    10.0.0.0/8 is directly connected, Serial0/3/0
    L    10.0.0.1/32 is directly connected, Serial0/3/0
    C    10.0.0.2/32 is directly connected, Serial0/3/0
    C    200.100.50.0/24 is variably subnetted, 2 subnets, 2 masks
    C    200.100.50.0/24 is directly connected, FastEthernet0/0
    L    200.100.50.1/32 is directly connected, FastEthernet0/0
    S    200.100.100.0/24 [1/0] via 10.0.0.2
```

Router-A#

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Router – B

```
Router-B(config)#ip route 200.100.50.0 255.255.255.0 10.0.0.1
Router-B(config)#exit
Router-B#
%SYS-5-CONFIG_I: Configured from console by console

Router-B#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
    C    10.0.0.0/8 is directly connected, Serial0/3/0
    C    10.0.0.1/32 is directly connected, Serial0/3/0
    L    10.0.0.2/32 is directly connected, Serial0/3/0
    S    200.100.50.0/24 [1/0] via 10.0.0.1
    C    200.100.100.0/24 is variably subnetted, 2 subnets, 2 masks
    C    200.100.100.0/24 is directly connected, FastEthernet0/0
    L    200.100.100.1/32 is directly connected, FastEthernet0/0
```

Router-B#

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Telnet

I configured Telnet on both routers as instructed and verified the setup by telnetting into the other router.

Router – B

```
Router-B#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-B(config)#enable password Enable123
Router-B(config)#line vty 0 4
Router-B(config-line)#password line123
Router-B(config-line)#login
Router-B(config-line)#
```

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Router – A

```
Router-A#telnet 10.0.0.2
Trying 10.0.0.2 ...Open

User Access Verification

Password:
Router-B>en
Password:
Router-B#show ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          200.100.100.1   YES manual up          up
FastEthernet0/1          unassigned      YES unset   administratively down down
Serial0/3/0              10.0.0.2        YES manual up          up
Vlan1                    unassigned      YES unset   administratively down down
Router-B#
```

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Configuration with Local User

I created two users on Router A: user1 and user2, with passwords set as user123 and user1234, respectively. I verified the existence of these users, noting that only the user with a plain password is displayed in full, while the user with a secret password is shown with its password in a hashed format. Additionally, I configured Telnet so that when another router attempts to connect to Router A, it prompts for a username and the corresponding password for login. I also set a password to ensure that only those who know it can access privilege mode.

Router – A

Local User Database:

```
Router-A#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-A(config)#username user1 password user123
Router-A(config)#username user2 secret user1234
Router-A(config)#exit
Router-A#
```

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Verification:

```
Router-A#  
%SYS-5-CONFIG_I: Configured from console by console  
  
Router-A#show running-config | include username  
username user1 password 0 user123  
username user2 secret 5 $1$mERr$SND.5vWo0rJ1yZz3NkaQn.  
Router-A#
```

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Telnet Configuration:

```
Router-A#config t  
Enter configuration commands, one per line. End with CNTL/Z.  
Router-A(config)#line vty 0 4  
Router-A(config-line)#no password  
Router-A(config-line)#transport input telnet  
Router-A(config-line)#login local  
Router-A(config-line)#exit  
Router-A(config)#enable password Enable12  
Router-A(config)#
```

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Telnet Router A from Router B

I used Telnet to connect to Router A from Router B and logged in as user1 using the password user123. After logging in, I attempted to access privilege mode on Router A, which prompted me to enter a password. Once I entered the password, I briefly checked the interfaces.

Router – B

```
Router-B#telnet 10.0.0.1  
Trying 10.0.0.1 ...Open  
  
User Access Verification  
  
Username: user1  
Password:  
Router-A>en  
Password:  
Router-A#show ip int brief  


| Interface       | IP-Address   | OK? | Method | Status                | Protocol |
|-----------------|--------------|-----|--------|-----------------------|----------|
| FastEthernet0/0 | 200.100.50.1 | YES | manual | up                    | up       |
| FastEthernet0/1 | unassigned   | YES | unset  | administratively down | down     |
| Serial0/3/0     | 10.0.0.1     | YES | manual | up                    | up       |
| Vlan1           | unassigned   | YES | unset  | administratively down | down     |

  
Router-A#
```

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Router – A

I set up two passwords in privilege mode: one as plain text that anyone can see, and the other is hashed, which will be displayed whenever a user attempts to view the running configuration.

```
Router-A(config)#enable password Enable12
Router-A(config)#enable secret Enable12345
Router-A(config)#
```

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Router – B

On Router B, I accessed Router A using Telnet and logged in with the username "user2" and the password "user1234." After successfully logging in, I accessed privileged mode using the two passwords we had set up earlier. I then entered the configuration terminal and removed both passwords remotely from Router B via Telnet. When I attempted to access privileged mode after removing the passwords, I encountered an error indicating that the password was not set up.

```
Router-B#telnet 10.0.0.1
Trying 10.0.0.1 ...Open

User Access Verification

Username: User2
Password:
% Login invalid

Username: user2
Password:
Router-A>en
Password:
Password:
Router-A#configuration
Translating "configuration"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router-A#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-A(config)#no enable password
Router-A(config)#no enable secret
Router-A(config)#exit
Router-A#exit
```

```
[Connection to 10.0.0.1 closed by foreign host]
Router-B#telnet 10.0.0.1
Trying 10.0.0.1 ...Open
```

```
User Access Verification
```

```
Username: user2
Password:
% Login invalid
```

```
Username: user2
Password:
Router-A>en
% No password set.
Router-A>en
% No password set.
Router-A>
```

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Router – A

Came back to router A to set up new passwords that were previously removed from router B using Telnet.

```
Router-A(config)#enable password Enable123
Router-A(config)#enable secret secretEnable
Router-A(config)#exit
Router-A#
%SYS-5-CONFIG_I: Configured from console by console

Router-A#
%SYS-5-CONFIG_I: Configured from console by console

Router-A#
```

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Router – B

I accessed the privilege mode of router A from router B using the secret password. This allowed me to bypass the need to enter the plain password successfully. After that, I removed the privilege mode passwords and opened the configuration terminal. By using the command **`show running-config`**, I was able to view the complete configuration of router A, including usernames and their corresponding passwords. However, the password that was set as a secret was displayed in a hashed format.

```
Router-B#telnet 10.0.0.1
Trying 10.0.0.1 ...Open

User Access Verification

Username: user2
Password:
Router-A>en
Password:
Router-A#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-A(config)#no enable password
Router-A(config)#exit
Router-A#exit

[Connection to 10.0.0.1 closed by foreign host]
Router-B#
```

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```
Router-B#telnet 10.0.0.1
Trying 10.0.0.1 ...Open

User Access Verification

Username: user2
Password:
Router-A>en
Password:
Password:
Router-A#show running-config
Building configuration...

Current configuration : 946 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router-A
!
!
!
enable secret 5 $1$mERr$sbldbMWh7ntiEgHS8PjgN/
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
username user1 password 0 user123
username user2 secret 5 $1$mERr$SSND.5vWo0rJ1yZz3NkaQn.
!
!
```

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```
interface FastEthernet0/0
ip address 200.100.50.1 255.255.255.0
duplex auto
speed auto
!
interface FastEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/3/0
bandwidth 1000
ip address 10.0.0.1 255.0.0.0
encapsulation ppp
clock rate 64000
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 200.100.100.0 255.255.255.0 10.0.0.2
!
ip flow-export version 9
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login local
transport input telnet
!
!
!
end
```

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Remaining portion of Lab Task #02

Blocking the incoming SSH Traffic:

Applied the rule on incoming ssh traffic requesting for remote access.

```
(root@kali)~# iptables -I INPUT -s 192.168.1.120 -p tcp --dport 22 -j DROP

(root@kali)~# iptables --list
Chain INPUT (policy ACCEPT)
target     prot opt source                destination
DROP      tcp  --  192.168.1.120          anywhere            tcp dpt:ssh

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

(root@kali)~#
```

Verifying from the Host:

SSH connection refused from the host machine trying to access the Linux.

```
C:\Users\HP>ssh -vvv 192.168.180.129
OpenSSH_for_Windows_9.5p1, LibreSSL 3.8.2
debug3: Failed to open file:C:/Users/HP/.ssh/config error:2
debug3: Failed to open file:C:/ProgramData/ssh/ssh_config error:2
debug2: resolve_canonicalize: hostname 192.168.180.129 is address
debug3: expanded UserKnownHostsFile '~/.ssh/known_hosts' -> 'C:\\Users\\HP/.ssh/known_hosts'
debug3: expanded UserKnownHostsFile '~/.ssh/known_hosts2' -> 'C:\\Users\\HP/.ssh/known_hosts2'
debug3: ssh_connect_direct: entering
debug1: Connecting to 192.168.180.129 [192.168.180.129] port 22.
debug3: finish_connect - ERROR: async io completed with error: 10061, io:00000239CADB1630
debug1: connect to address 192.168.180.129 port 22: Connection refused
ssh: connect to host 192.168.180.129 port 22: Connection refused

C:\Users\HP>
```

Blocking the outgoing SSH traffic:

Removed the previously applied incoming rule and then applied the outgoing SSH traffic rule.

```
(root@kali)~# iptables -F

(root@kali)~# iptables --list
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

(root@kali)~# iptables -I OUTPUT -s 192.168.1.120 -p tcp --dport 22 -j DROP

(root@kali)~# iptables --list
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
DROP      tcp  --  192.168.1.120          anywhere            tcp dpt:ssh

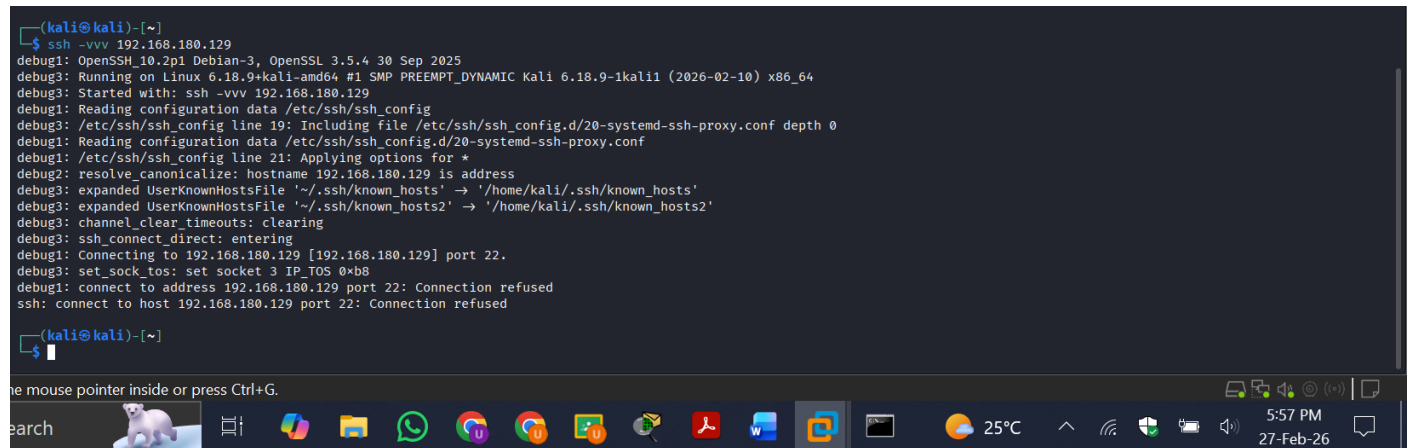
(root@kali)~#
```

Verifying:

Verified the applied rule by requesting the host from the VM.

```
(kali@kali) [~]
$ ssh -vvv 192.168.180.129
debug1: OpenSSH_10.2p1 Debian-3, OpenSSL 3.5.4 30 Sep 2025
debug3: Running on linux 6.18.0-kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.9-1kali1 (2026-02-10) x86_64
debug3: Started with: ssh -vvv 192.168.180.129
debug1: Reading configuration data /etc/ssh/ssh_config
debug3: /etc/ssh/ssh_config line 19: Including file /etc/ssh/ssh_config.d/20-systemd-ssh-proxy.conf depth 0
debug1: Reading configuration data /etc/ssh/ssh_config.d/20-systemd-ssh-proxy.conf
debug1: /etc/ssh/ssh_config line 21: Applying options for *
debug2: resolve_canonicalize: hostname 192.168.180.129 is address
debug3: expanded UserKnownHostsFile '~/.ssh/known_hosts' -> '/home/kali/.ssh/known_hosts'
debug3: expanded UserKnownHostsFile '~/.ssh/known_hosts2' -> '/home/kali/.ssh/known_hosts2'
debug3: channel_clear_timeouts: clearing
debug3: ssh_connect_direct: entering
debug1: Connecting to 192.168.180.129 [192.168.180.129] port 22.
debug3: set_sock_tos: set socket 3 IP_TOS 0xb8
debug1: connect to address 192.168.180.129 port 22: Connection refused
ssh: connect to host 192.168.180.129 port 22: Connection refused

(kali@kali) [~]
$
```



Conclusion:

Setting up an SSH connection needs data to flow both ways. Even if you're just planning to send commands, the protocol has to do a cryptographic handshake and acknowledge received packets. If you have a rule that only allows incoming traffic, the server won't be able to send back the necessary "Server Hello" or authentication challenges to the client. On the flip side, a rule that only allows outgoing traffic stops the server from getting the initial connection request (SYN packet) from the client, causing the connection to time out before it even starts.

THE END